

Report AI Project



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AI



Introduction

The **8-Queens Problem** is one of the most well-known classical problems in **Artificial Intelligence** and **Search Algorithms**. The objective is to place **eight queens** on an **8×8 chessboard** such that no two queens can attack each other.

This means that queens must not share the same:

Row

Column

Diagonal (left or right)

In this project, **six different algorithms** were implemented to solve the **8-Queens problem**, allowing a comparison between various search and optimization strategies.

State Representation

All algorithms use the same state representation:

state = [3, 1, 6, 4, 0, 7, 5, 2]

Index = Column number (0 → 7)

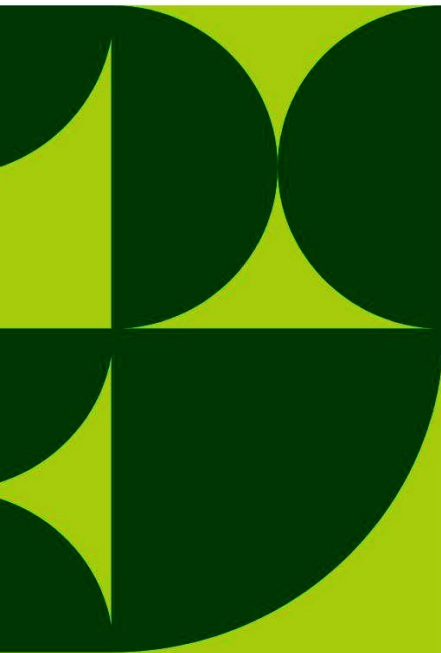
Value = Row number where the queen is placed in that column

Example:

`state[0] = 3` → the queen in column 0 is placed in row 3.

This representation guarantees one queen per column, therefore reducing the search space from $64 \times 64 \times 64 \times 64 \times 64 \times 64 \times 64 \times 64$ to $8 \times 8 \times 8 \times 8 \times 8 \times 8 \times 8 \times 8$.





Algorithms Used

1 Depth-Limited Search (**Backtracking**)

Type: **Systematic Search**

Concept:

Place one queen per column, left to right.

For each column, try all rows (0–7) and backtrack whenever a conflict is found.

When column 8 is reached → a valid solution is found.

Advantages

Complete (guaranteed to find a solution if it exists)

Simple and easy to implement

Disadvantages

Can be slow for large N

Expected performance: < 0.1 seconds

2 Best-First Search

Type: **Informed Greedy Search**

Concept:

At each step, select the column with the **highest number of conflicts**, and move its queen to the **row that minimizes attacks**.

Advantages

Smarter than basic backtracking

Focuses directly on fixing conflicts

Disadvantages

May get stuck in **local optima**

Not always guaranteed to reach a solution

Expected performance: Fast, but not always successful



3 Hill Climbing (Min-Conflicts) ★

Type: Local Optimization Algorithm

Most effective algorithm for N-Queens

Concept:

Start with a random state (one queen per column)

Select a conflicting column randomly

Move the queen to the row with **minimum conflicts**

Repeat until conflicts = 0

Advantages

Extremely fast (< 0.05 seconds)

Very efficient for N-Queens

Widely used in practical scheduling and planning problems

Expected performance: Fastest algorithm in the project

4 Minimax (Limited Depth)

Type: Adversarial Search

(Used here only for educational purposes)

Concept:

Assumes two imaginary players:

One tries to **reduce** conflicts

The other tries to **increase** conflicts

Minimax selects the best move assuming an optimal opponent.

Issues

There is no real opponent in this problem

Depth was limited to 6 → cannot find a full solution

Very slow

Evaluation: Interesting experiment, but not practical

5 Alpha-Beta Pruning

Improved version of Minimax by eliminating unnecessary branches (pruning)

Advantages

Faster than basic Minimax

Reduces the number of expanded nodes

Disadvantages

Still not suitable for the 8-Queens problem

Not competitive with real search methods here

Result: Slight improvement over Minimax, but still inefficient

6 Genetic Algorithm

Type: Evolutionary Search

Concept:

Start with a **population** of random states

Evaluate **fitness** = number of conflicts (lower is better)

Apply:

Selection (e.g., Tournament)

Crossover

Mutation

Continue evolving for many generations until a perfect state is found

Advantages

Powerful for large and complex problems

Does not require problem-specific knowledge

Disadvantages

Slower compared to Hill Climbing (seconds)

Performance depends on parameter tuning (population size, mutation rate, ...)

Expected performance: Finds a solution after 1000–2000 generations (< 5 seconds)

Performance Comparison (Approximate)

Rank	Algorithm	Speed	Solution Quality	Notes
1	Hill Climbing (Min-Conflicts)	★ Extremely Fast	Excellent	Best overall for N-Queens
2	Depth-Limited Search	Fast	Excellent	Complete and reliable
3	Genetic Algorithm	Medium	Very Good	Useful for larger N
4	Best-First Search	Fast	Good	May fail
5	Alpha-Beta Pruning	Slow	Weak	Not suitable
6	Minimax	Very Slow	Weak	Educational only

Conclusion & Recommendations

Algorithm When to Use

Min-Conflicts (Hill Climbing) Best choice for any N-Queens problem

Backtracking (DLS)	When N is small and a guaranteed solution is needed
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Genetic Algorithm	For larger or more complex versions like 100-Queens
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Best-First Search Good for learning heuristic search techniques

Minimax / Alpha-Beta	Only for competitive games (Chess, Tic-Tac-Toe, ...)
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Final Notes

This study clearly shows that success depends on:

The search strategy

The problem structure

The size of the state space

Min-Conflicts provides the best balance between speed and solution accuracy, making it the most suitable algorithm for the 8-Queens problem.